

EMBERGUARD · REQUIREMENTS

EmberGuard requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, reference house and status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

EmberGuard requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs. The status column gives the position of the concept in EGD-PRC-001 v0.2, based on first-order estimates that will be checked by calculation at TRL 3. Two requirements are **not met** (R6 on net wetting, R13 on cost), one is **partly met** (R2 on single embers) and one is **not met by the kit alone** (R8 on pump power).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Detect a smouldering ignition on a watched gutter or roof edge	A 100 cm ² (15.5 in ²) hot spot at 300 °C or hotter, anywhere along the watched gutters from 1.5 m to 13 m from the mast, within 10 s	Radiometric calculation from sensor field of view and noise; later a heated-target trial	Met on paper, large margin (estimate)
R2	Detect single landed embers	A 10 mm glowing ember at 600 °C or hotter within 8 m of the mast, within 10 s	Same as R1	Partly met: a 600 °C ember gives about 3 K of pixel rise at 8 m, below the proposed 5 K trigger threshold; it reaches the threshold only within about 5 m. An 800 °C ember is at the threshold at 8 m (estimate)
R3	Watch both roof planes	Both eave gutters of the reference house, each from 1.5 m from the gable end to the far end	Geometry of fields of view on the massing model	Met except the first 1.3 m of each gutter nearest the mast
R4	Arm only in fire weather	Automatic arming when sustained wind is 30 km/h (19 mph) or more, or gusts 50 km/h (31 mph) or more, with relative humidity 20 % or less for 10 min; manual and remote arming; defaults adjustable	Design review of the arming logic	Met by design
R5	Start water quickly	Water at the farthest head within 60 s of a confirmed detection	Line fill calculation	Met, about 55 s with empty lines (estimate)
R6	Wet the gutter and roof edge	Net water on a 1.2 m wide strip along each eave (gutter, fascia and first metre of roof) of 5 mm/h or more, averaged over each spray cycle, at the design wind of 30 km/h	Application-rate calculation with a drift allowance; later a spray trial in wind	Not met: about 7.7 mm/h gross but about 3.8 mm/h net at an assumed 50 % wind drift (estimate)
R7	Use little water	Steady draw 4 L/min (1.1 US gal/min) or less; 1,000 L (264 US gal) or less for a 4 h ember event	Flow calculation from head ratings and duty cycle	Met, 4 L/min and about 960 L; no margin
R8	Work through a grid outage	Sensing, logic and valves: 72 h armed plus 4 h of spraying on the battery alone, with no solar input. Water supply must not depend on grid power	Energy budget	Kit part met, about 64 Wh needed against about 69 Wh usable. Not met by the kit alone where the pump runs on grid power
R9	Fail safely	Loss of sensor data while armed starts spraying; low battery, no water pressure or a faulty valve raises an alarm; power loss closes the valves	Failure modes review	Met by design
R10		Mast and head survive gusts of 120 km/h (75 mph); electronics operate	Wind load and thermal calculations	Wind met on first-order check (about 25 MPa bending stress in

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	Survive fire weather until the front arrives	from -10 to 60 °C ambient; head shaded from sun and radiant heat		the mast, estimate); thermal unverified
R11	Install without roof work or mains wiring	Mast, cable and spray lines fixed to walls, fascia and gutter lips with clamps; no roof penetrations; 12 V DC only; installable by two people with hand tools in 6 h or less	Design review; later a timed trial	Met by design; install time unverified
R12	Tell people what it is doing	Local siren and status light; phone alert when armed, spraying, faulted or low on battery when a network is available; event log of detections, wind and valve actions	Design review	Met by design; phone alerts need a working network
R13	Stay within the concept budget	Kit parts \$300 or less, excluding the water source and pump	Priced BOM	Not met: about \$420 (indicative)

■ Reference house and assumptions

- **House.** Single storey, 12 x 8 m (39 x 26 ft) footprint, 2.7 m walls, gable roof at 22 degrees with the ridge along the long side, 500 mm overhangs, gutters along both long eaves (13 m each, 26 m total). The sensor mast stands off one gable end at the ridge line.
- **Water.** A 1,000 L tank with a pump on its own supply, or a mains connection that holds at least 2 bar at 4 L/min. EmberGuard does not include the tank or pump.
- **Ember event.** 4 h of active spraying within a 72 h armed period. This is a design assumption, not a measured duration; real ember exposure varies from minutes to many hours.
- **Heads.** Six micro-sprinklers per eave at about 2.2 m spacing, about 40 L/h each at 2 bar (typical catalogue figure for small irrigation heads, to be confirmed with a named part).
- **Wind drift.** 50 % of the sprayed water misses the target strip at the design wind. This is an assumption with high uncertainty and is the main reason R6 is not met.
- **Ember targets.** Emissivity 0.9, sensor band 8 to 14 µm, background 300 K. See EGD-PRC-001 for the radiometric estimate.

Safety: EmberGuard does not replace evacuation. Requirements R4 to R9 describe behaviour while the house is empty. Nothing in these requirements permits anyone to stay behind to operate or watch the system.